

Workflow and architecture patterns for working together 1: Workflows

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Outline

1 Current challenges and their canonical solutions

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- 2 DevOps 201
 - Communication
 - Conway's Law
 - The Mythical Man Month
 - Throughput
 - Just-in-time manufacturing
 - Goldratt's theory of constraints
 - DevOps Research and Assessment (DORA)

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1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' *Datamation*, 1967

Conway's law: examples

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- front-end [devs] business layer [backend devs] monolithic database [DBA team]
- a web api controller [manager] delegates most business logic to business classes [developers] which are part of the same in-memory process [team], while serving as a single entry-point for wider cross-network [cross-team] communication.

Conway's law: more examples

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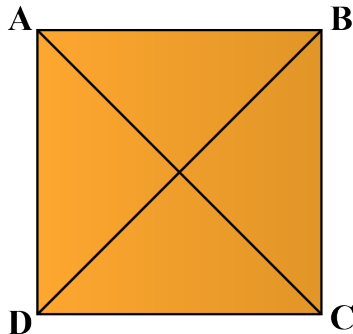
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Conway's law: corollaries

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 - branching-based strategies for team management compromise between the above approaches without their benefits

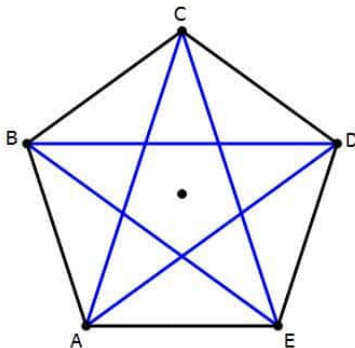
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $\frac{n(n-1)}{2}$



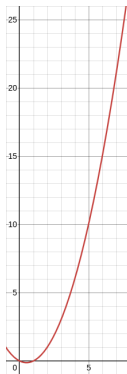
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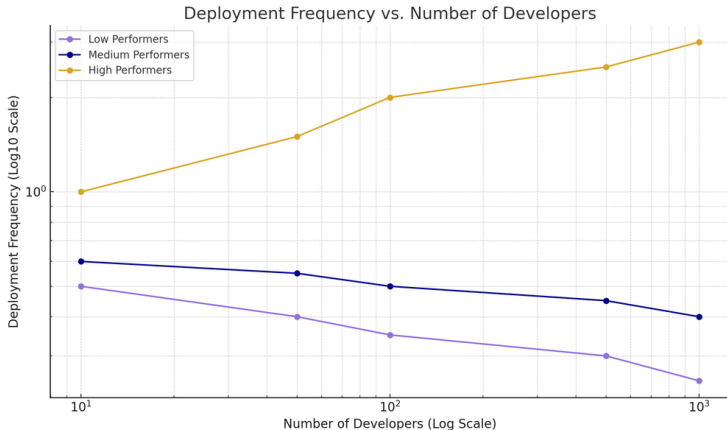
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The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



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Just-in-time manufacturing and Goldratt's theory of constraints

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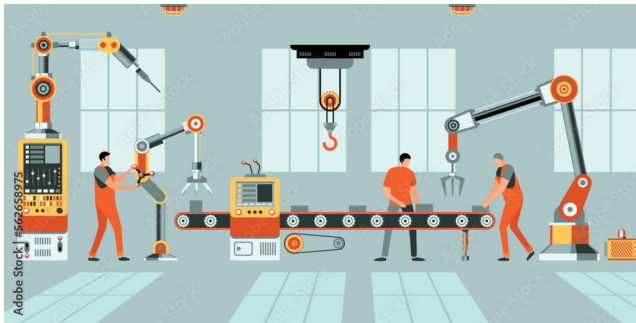
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Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
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- Remove bottlenecks

Goldratt's theory of constraints (cont.)



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- Led by a PhD statistician, Nicole Forsgren, from 2013-2019
- Uses *cluster analysis* to discover groupings - has no prior understanding of what counts as good or bad.

The State of DevOps Report: throughput metrics

- lead time for changes

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- lead time for changes
- deployment frequency

The State of DevOps Report: stability metrics

- change failure rate

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- change failure rate
- mean time to restore

The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, *Accelerate* (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x

faster lead
time

182x

more
deployments
per year

8x

lower change
failure rate

2293x

faster failed
deployment
recovery times

- 2024 State of DevOps Report

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- leads to bottlenecks
- delays communication
- discourages cross-functional collaboration
- punishes untracked work (e.g. thorough code reviews/dev testing, helping blocked colleagues)

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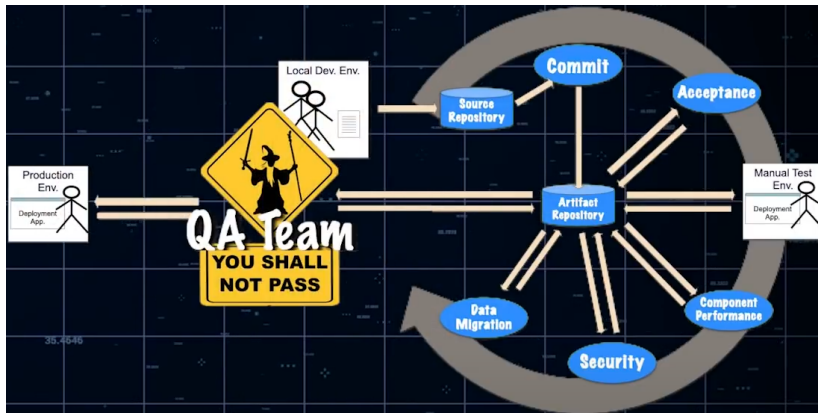
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- change approval boards
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- these all increase lead time

Throughput anti-pattern solutions: aggressively parallelize work

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- stop people from waiting on each other
- work in small batches
- That's mostly it

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'DevOps is whatever you do to bridge friction created by silos, and all the rest is engineering' - Patrick Debois, Puppet State of DevOps report 2021

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